Modular Forms

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1 Introduction

Definition 1.1

Define the upper half plane

$$\mathfrak{h} = \{\tau \in \mathbb{C} \mid \operatorname{im}(\tau) > 0\}$$

and the positive determinant group

$$\mathrm{GL}_2(\mathbb{R})^+ = \{g \in \mathrm{GL}_2(\mathbb{R}) \mid \det(g) > 0\}$$

and

$$\Gamma(1) = \mathrm{SL}_2(\mathbb{Z})$$

Lemma 1.2. $\mathrm{GL}_2(\mathbb{R})^+$ acts transitively on $\mathfrak{h}$ via Möbius transformations.

Proof. Suppose $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{R})^+$, and $\tau \in \mathfrak{h}$. Then

$$\operatorname{Im}(g\tau) = \frac{1}{2i}\left(\frac{a\tau+b}{c\tau+d} - \frac{a\overline{\tau}+b}{c\overline{\tau}+d}\right) = \frac{1}{2i}\frac{(ad-bc)(\tau-\overline{\tau})}{|c\tau+d|^2} = \frac{\deg(g)\operatorname{Im}(\tau)}{|c\tau+d|^2} > 0$$

For transitivity, for $x + iy \in \mathfrak{h}$,

$$\begin{pmatrix} y & x \\ 0 & 1 \end{pmatrix} \cdot i = x + iy$$

□

Definition 1.3

Let $g \in \mathrm{GL}_2(\mathbb{R})^+$, $\tau \in \mathfrak{h}$, define $j(g,\tau) = c\tau + d$, where $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. This is called the *modular cocycle*. If k is an integer, $f : \mathfrak{h} \to \mathbb{C}$ is any function, then

$$\begin{aligned} f|_k[g] : \mathfrak{h} &\to \mathbb{C} \\ \tau &\mapsto \det(g)^{k-1} f(g\tau) j(g,\tau) \end{aligned}$$

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is called the *weight k action of g on f*.

Lemma 1.4. This is a *right* action of $\mathrm{GL}_2(\mathbb{R})^+$. That is,

$$f|_k[gh] = (f|_k[g])\,|_k[h]$$

Proof.

$$\begin{aligned} f|_k[g]|_k[h](\tau) &= \det(h)^{k-1} f|_k[g](h\tau) j(h,\tau)^{-k} = \det(h)^{k-1}\det(g)^{k-1} f(gh\tau) j(g,h\tau)^{-k} j(h,\tau)^{-k} \\ &\overset{?}{=} \det(gh)^{k-1} f(gh\tau) j(gh,\tau)^{-k} \end{aligned}$$

Therefore, suffices to show that

$$j(gh,\tau) = j(g,h\tau)j(h,\tau)$$

Note that if $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $g\begin{pmatrix}\tau\\1\end{pmatrix} = \begin{pmatrix} a\tau+b \\ c\tau+d\end{pmatrix} = j(g,\tau)\begin{pmatrix} g\tau \\ 1\end{pmatrix}$, This means that,

$$j(gh,\tau)\begin{pmatrix} gh\tau \\ 1\end{pmatrix} = gh\begin{pmatrix}\tau\\1\end{pmatrix} = g\left(j(h,\tau)\begin{pmatrix} h\tau\\1\end{pmatrix}\right) = j(g,h\tau)j(h,\tau)\begin{pmatrix} gh\tau\\1\end{pmatrix}$$

□

Formulae:
For $g, h \in \mathrm{GL}_2(\mathbb{R})^+$, $\tau \in \mathfrak{h}$,

$$\mathrm{Im}(g\tau) = \frac{\det(g)\,\mathrm{Im}(\tau)}{|j(g,\tau)|^2}$$

and

$$j(gh,\tau) = j(g,h\tau)j(h,\tau)$$

Definition 1.5

Let $k \in \mathbb{Z}$, $\Gamma \leq \Gamma(1)$ a finite index subgroup. A *weakly modular function of weight k and level Γ* is a meromorphic function $f : \mathfrak{h} \to \mathbb{C}$, which is invariant under the weight k action of Γ. That is, such that for all $\gamma \in \Gamma$,

$$f|_k[\gamma] = f$$

We will define a modular form (when $\Gamma = \Gamma(1)$) next time, but they are weakly modular functions, which are holomorphic in $\mathfrak{h}$ and at ∞.

In fact, modular forms of fixed weight and level live in finite dimensional $\mathbb{C}$-vector spaces $M_k(\Gamma)$, which are the main objects of study in this course.

Why do we study modular forms?

1. They are related to the theory of elliptic functions. Let $E/\mathbb{C}$ be an elliptic curve. Let ω be a non-zero holomorphic 1-form. Then there exists a unique lattice $\Lambda \leq \mathbb{C}$, and an isomorphism of Riemann surfaces, $\phi : \mathbb{C}/\Lambda \to E$, such that $\phi^*(\omega) = \mathrm{d}z$. We can show that E is isomorphic to the elliptic curve $y^2 = 4x^3 - 60G_4(\Lambda)x - 140G_6(\Lambda)$, where for $k \in \mathbb{Z}$,

$$G_k(\Lambda) = \sum_{\lambda \in \Lambda\setminus 0} \lambda^{-k}$$

which converges for $\lambda > 2$. If $\tau \in \mathfrak{h}$, then we have an associated lattice $\Lambda_\tau = \mathbb{Z} \oplus \mathbb{Z}_\tau$, and the function

$$G_k(\tau) = G_k(\Lambda_\tau)$$

is a modular form of weight k and level $\Gamma(1)$. This is called the *Eisenstein series*. Moreover, $\mathfrak{h}/\mathrm{SL}_2(\mathbb{Z})$ can be identified (as a set) with the set of isomorphism classes of elliptic curves over $\mathbb{C}$.

2. Modular forms f have Fourier expansions
$$\sum_{n\in\mathbb{Z}} a_n q^n$$
where $a_n \in \mathbb{C}$, and often serve as generating functions for arithmetically interesting sequences a_n. One example is
$$\vartheta(q) = \sum_n e^{\pi i n^2 \tau}$$
If $k \in 2\mathbb{N}$, then ϑ^{2k} is a modular form, and on the other hand,
$$\vartheta^{2k} = \sum_{n\in\mathbb{Z}} r_k(n) e^{\pi i n \tau}$$
where $r_k(n)$ is the number of ways to write n as a sum of k squares. By expressing ϑ^{2k} in terms of ther modular forms, we can prove formulae such as
$$r_4(n) = 8 \sum_{d|n, 4\nmid d} d$$

3. The Riemann ζ function is an important object in number theory. Properties include
   (a) The Euler product
   $$\zeta(s) = \prod_p (1 - p^{-s})^{-1}$$
   (b) A meromorphic extension to $\mathbb{C}$,
   (c) A functional equation relating $\zeta(s)$ and $\zeta(1-s)$.

   A *Dirichlet L-series* is a function of the form
   $$\sum_{n\geq 1} a_n n^{-s}$$
   which has similar properties is called an L-function. Modular forms can be used to construct interesting examples of L-functions. Take $M_k(\Gamma)$ and decompose them under the action of Hecke operators. In particular, if $\Gamma = \Gamma(1)$, we get a decomposition into lines, called Hecke eigenforms.

4. Connections to the Langland programme, which predicts (among other things) a relation between modular forms and other objects in arithmetic geometry. A special case of this is the *Modularity conjecture*, which says that there is a bijection between elliptic curves over $\mathbb{Q}$ (up to isogeny) and the set of Hecke eigenforms of weight 2. This implies Fermat's last theorem. The bijection is formulated in the language of Hecke operators and L-functions.

   Homework: Handout on Moodle called "Reminder of Complex Analysis".

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